AES (Advanced Encryption Standard) Encryption Flowchart

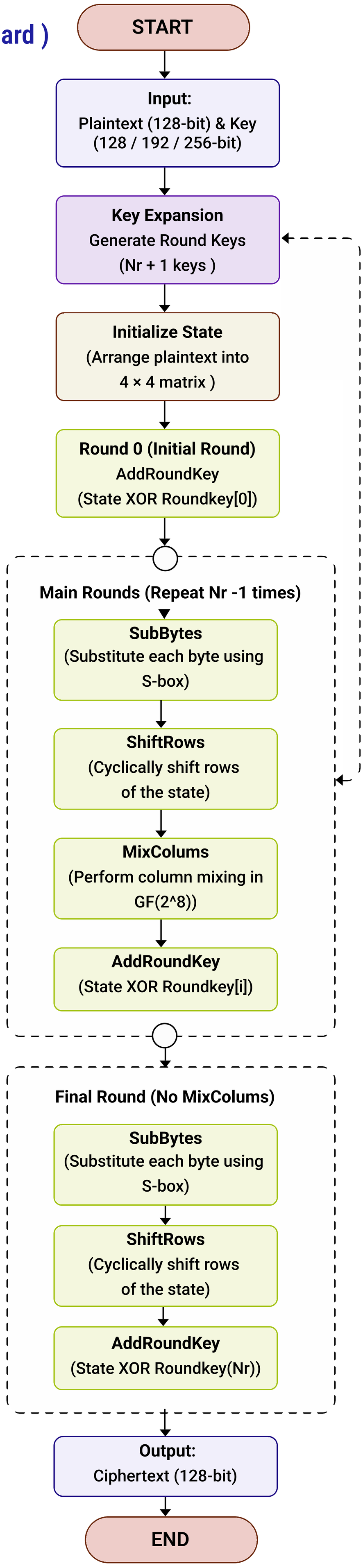
- Input / Output
- Process
- Transformation Step
- Key Schedule
- Decision
- Loop / Iteration

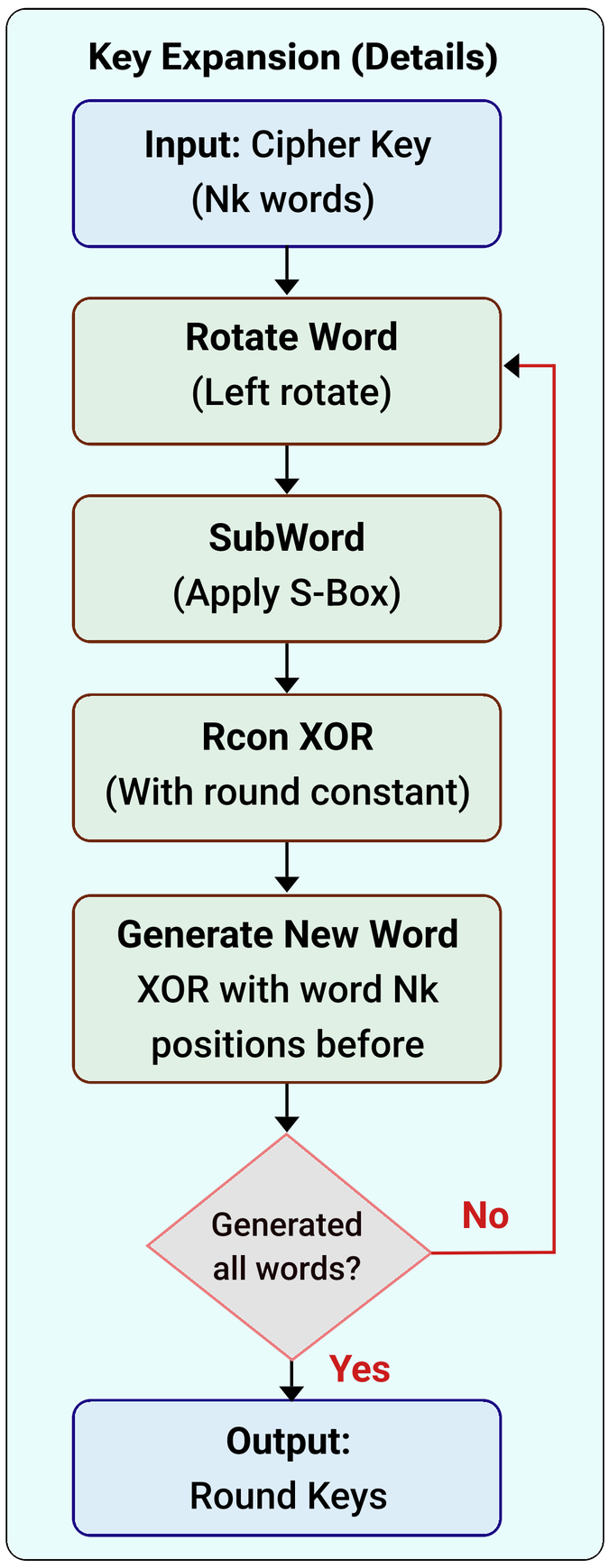
Number of Rounds (Nr)

- 128-bit Key → 10 rounds
- 192-bit Key → 12 rounds
- 256-bit Key → 14 rounds

Brief Description of Steps

- **AddRoundKey:** XOR each byte of the state with the round key.
- **SubBytes**: Non-linear byte substitution using S-Box.
- **ShiftRows**: Shift rows of the state to the left.
- **MixColumns**: Mix the data within each column of the state.





Notation

- State: 4×4 byte matrix
- Nb = 4 (block size)
- Nk = key length / 32
  (4, 6, 8 words)
- Nr = number of rounds
  (10, 12, 14)

State Representation (4×4)

$S_{0,0}$	$S_{0,1}$	$S_{0,2}$	$S_{0,3}$
$S_{1,0}$	$S_{1,1}$	$S_{1,2}$	$S_{1,3}$
$S_{2,0}$	$S_{2,1}$	$S_{2,2}$	$S_{2,3}$
$S_{3,0}$	$S_{3,1}$	$S_{3,2}$	$S_{3,3}$